T7.2.3-Worksheet 2-Pseudocode-Pre and Post-Conditional Loop

Answer the following questions.

1. Write pseudocode using a pre-conditional loop that repeatedly prompts the user to enter positive integers and calculates their sum. The input sequence terminates when the user enters `-1` (a sentinel value). Output the total sum after the loop ends. Declare all variables used.

2. A student wants to save money until their total savings reach at least **$100**. Write pseudocode using a `WHILE` loop that begins with a total savings of 0, repeatedly asks the user to enter an amount saved, and adds it to the total. The loop must stop as soon as the total reaches or exceeds $100. Output the final total saved. Declare all variables used.

3. Write pseudocode using a post-conditional loop that prompts the user to enter an exam score. The score must be between **0 and 100** inclusive. If the user enters a value outside this range, the program must re-prompt them. Once a valid score is entered, output `"Score accepted"`. Declare all variables used.

4. A secure lock program requires the user to enter a 4-digit PIN. The correct PIN is `1234`. Write pseudocode using a `REPEAT...UNTIL` loop that repeatedly prompts the user to enter the PIN until the correct one is entered. When the correct PIN is provided, display `"Access granted"`. Declare all variables used.

5. A laboratory cooler needs to lower the temperature of a sample until it drops to **0°C or below**. The starting temperature is entered by the user. If the temperature is already 0°C or colder at the start, the cooling process must not execute at all. Otherwise, the program repeatedly simulates cooling by decreasing the temperature by **2°C** per cooling cycle and displaying the new temperature after each cycle. Once the target is reached, output `"Cooling process complete"`. Write an algorithm in pseudocode to solve this problem. Declare all variables used.

6. Write an algorithm in pseudocode to calculate the area of rectangles for multiple rooms.
    a. The program must prompt the user to input the `Length` in meters.
    b. A `Length` value of `0` indicates that there are no rooms to process and must immediately stop the program without requesting any further inputs or performing any calculations.
    c. For any valid `Length` greater than `0`, input the `Width` in meters.
    d. Calculate the area using $Area = Length \times Width$ and output the calculated area.
    e. The process must continue asking for room dimensions until the user enters 0 for the `Length`.
    f. When finished, output the message "`All room areas calculated`".
    g. Declare all variables used with appropriate data types.

7. A bus reservation system requires passengers to choose a seat number from **1 to 40**.

    Write an algorithm in pseudocode that:

    - Prompts the passenger to enter their desired seat number.
    - Checks if the number is within the valid range (1 to 40 inclusive).
    - Continues to re-prompt the user with an error message "`Invalid seat number, try again`" as long as the input is invalid.
    - Terminates only when a valid seat number is entered and outputs "`Seat number confirmed:`" followed by the chosen seat.

    Declare all variables used with standard IGCSE data types.

End of Question Paper